UAV-based Search and Rescue: Human Detection Using Convolutional Neural Networks

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Abstract. Unmanned Aerial Vehicles (UAVs) have proven to be valuable tools for search and rescue (SAR) operations, enabling rapid aerial surveillance over large and inaccessible areas. Using recent advances in deep learning, this study explored the application of Convolutional Neural Networks (CNNs) to detect human presence in aerial images. The images were preprocessed, divided into smaller patches, and augmented to enhance the robustness of the model. The performance of the multiple CNN architectures was evaluated to determine the most effective approach. The results showed a very good capability of the CNN models tested when detecting humans, achieving AU-ROC values of 0.95. In addition, an interactive web application was developed using StreamLit, allowing users to upload images and test different pre-trained models, thus providing a practical tool for SAR teams.

Keywords: Human Detection · Convolutional Neural Networks · Search and Rescue · Aerial Images

1 Introduction

The HERIDAL dataset was used in this study, which consists of high-resolution aerial images captured over various terrain [1] [4]. The dataset includes both real and synthetically generated image patches containing human and negative samples without human presence. Previous studies successfully applied CNNs to detect humans in aerial images from this dataset [2]. Images were collected using UAVs at different altitudes and lighting conditions, to enhance the robustness of the model.

All images in the dataset were labeled using an XML annotation format following the Pascal VOC standard. For images containing humans, each person is annotated with a bounding box defined by its pixel coordinates.

$$(x_{min}, y_{min}, x_{max}, y_{max})$$

These annotations are essential for supervised learning because they provide the precise localization of human instances within images.

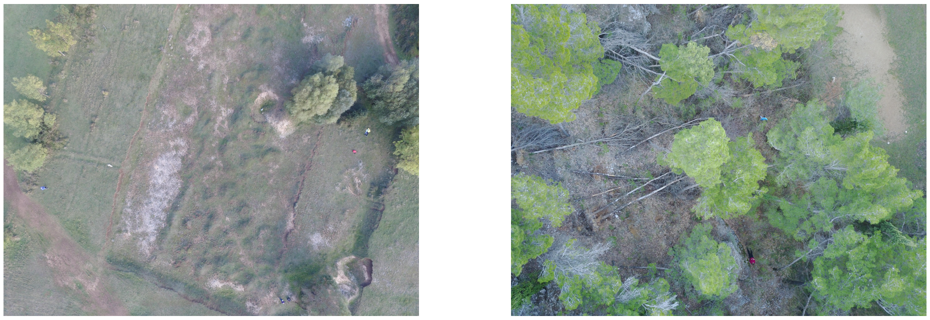

Fig. 1: Example images from the HERIDAL dataset used for person detection in natural environments.

2 Methodology

2.1 Preprocessing

The images were processed using a technique called tiling to facilitate human detection and improve training efficiency. This method involves splitting large images into smaller overlapping patches, in this case, of 340×340 pixels. The motivation behind this approach is to resize human figures to a scale that is suitable for classification networks.

The tiling process was performed with an overlap in both axes to ensure that no human instances were partially cut between patches. After the tiling step, the bounding box coordinates are re-mapped from the original images to the newly generated sub-images. This step preserves the accuracy of human localization, ensuring that the model receives the correctly labeled data during training.

2.2 CNN Architecture

The three CNN architectures used in this study are based on the AlexNet [3] structure, with specific variations in the number of dense layers and regularization parameters. These models are referred to as CNN-1, CNN-2, and CNN-3.

All three architectures share the following common characteristics.

- 5 convolutional layers
- 3 MaxPooling layers
- 4 normalization layers
- Learning rate of 0.001
- 100 epochs
- Early stopping with a patience of 5 epochs
- Batch sizes: 32 for training, 256 for validation, and 256 for testing

The main differences between the models lie in the number of dense layers, neurons, and dropout rates, as detailed in Table 1.

This configuration was designed to optimize the performance of the image classification tasks and evaluate the accuracy, AU-PR, and AUROC metrics.

Table 1: Comparison of CNN Architectures

Model	Dense Layers	Dropout	Neurons per Layer
CNN-1	1	0.2	1024
CNN-2	2	0.2 (first), 0.1 (second)	1024, 256
CNN-3	3	0.4 each layer	4096, 4096, 1000

2.3 Training

To evaluate model performance and ensure robustness, 5-fold cross-validation was applied while preserving the class and terrain distributions. Data augmentation techniques, such as flipping, rotation, and shifting, have been used to enhance generalization. To address the class imbalance, weights were assigned to each class based on their inverse frequencies. Additionally, multiple decision thresholds were considered to balance false positives and false negatives, providing flexibility in search and rescue operations based on the mission requirements.

3 Results and Web Application

The evaluation of the CNN models was based on accuracy, AU-PR, and AU-ROC metrics. The best-performing model, a modified AlexNet architecture with enhanced dense layers, achieved an AU-ROC score of 0.954, thereby demonstrating a high reliability in human detection. Table 3 provides a comparative analysis of the different evaluated CNN architectures.

Model	TN	FN	TP	FP	Accuracy	AU-PR	AU-ROC	Time (mins)
CNN1	14347	282	556	675	0.940	0.548	0.948	85.208
CNN2	13894	136	702	1128	0.920	0.552	0.954	80.212
CNN3	14513	322	516	509	0.948	0.536	0.937	102.355

Table 2: Performance metrics obtained for the models analyzed through cross-validation

A web-based tool was developed using StreamLit [1], which enables users to test pre-trained models on uploaded images. The application utilizes a user-friendly interface built using Python and StreamLit interactive widgets. Users can upload aerial images in common formats (e.g., JPEG and PNG) and select them from available pre-trained models. The application allows adjustment of prediction thresholds via a slider, enabling users to fine-tune the sensitivity of detection. The detection results were displayed as bounding boxes overlaid on the input image, and users exported the results as CSV files for further analysis.

[1] `https://streamlit.io/`

Figure 2 shows an example of model prediction using a web application. The red squares indicate sub-images classified as containing humans.

Fig. 2: Example of model prediction using the web application. Red bounding boxes indicate detected humans.

4 Conclusion

This study highlighted the potential of CNNs for UAV-based SAR operations. Future research will focus on optimizing model architectures and expanding the dataset to improve detection accuracy across diverse environments.

References

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